An Interactive Visualization Tool to Explore People's Tweets towards COVID-19

Shah Rukh Humayoun
San Francisco State University
San Francisco, California, USA
humayoun@sfsu.edu

Siddhita Chimote
San Francisco State University
San Francisco, California, USA
schimote@mail.sfsu.edu

ABSTRACT

The Twitter has become a powerful medium for people to express their feelings and reactions towards some recent event or happening event. Analyzing and exploring such tweet data gives us in-depth understanding of the underlying event from people perspectives. In this work, we use the power of visualization to explore people's tweets regarding the COVID-19 pandemic from the time period between April 2020 and March 2021. We use a number of visualizations targeting to show the sentimental polarity of people's tweets over time and important keywords in people's tweets and their evolution over time. The resulting visualizations give us an opportunity to explore people's feelings and reactions during a whole pandemic year and to see how it evolved over time.

CCS CONCEPTS

• **Human-centered computing** → **Graphical user interfaces**; **Information visualization**.

KEYWORDS

Information visualization, social media, tweet data, COVID-19.

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1 INTRODUCTION

Social media platforms (e.g., Twitter, Facebook, etc.) have become powerful medium for people to express their feelings and reactions towards some happened or currently happening events. Twitter is one of the prominent platforms where people use short tweets (previously 140 maximum characters and currently maximum 280 characters) to write their feelings and reactions in a precise way. This makes it an excellent medium to understand people's feelings and reactions towards a particular event compared to analyzing the same event on other social media platforms.

Keeping this in view, researchers have developed visualization tools to explore and understand people tweets from different perspectives. For example, many visualization tool focused on showing the sentimental analysis of tweets (e.g., [1][5] [4][7][11][12][15][16] [20]), while other visualization tools focus on analyzing and exploring tweet data based on the geo-spatial information (e.g., [3][8][13] [17] [18]). Some researchers focus on exploring topics and analyzing their evolution over time (e.g., [2][6][9][10][14][19]).

The COVID-19 outbreak was first announced in the news during the month of December 2019. However, the pandemic really began to take its toll in the early 2020, when it started gathering much more attention and people realized the severity nature of pandemic. As the lockdown started across the world due to the spread of pandemic, people started using more and more social media platforms to express their feelings and reactions towards COVID-19.

In this work, we use interactive visualizations to explore people's tweets regarding COVID-19, from the time period between April 2020 to March 2021, in order to see how people feelings and reactions towards COVID-19 evolved over the period of one year. We use a number of interactive visualizations targeting to show the sentimental polarity of people's tweets over time, important keywords in people's tweets and their evolution over time. The resulting visualizations give us an opportunity to explore people's feelings and reactions during a whole pandemic year and to see how it evolved over time. They also give us an indication how people's reactions and feeling changed or evolved during some peaks of COVID-19 in their areas.

2 THE DATA SETS

We collected the people's tweet data through two data sources. The first data source was George Washington University's TweetSets repository[1] for the period between April 01, 2020 to December 04, 2021. For collecting tweet data from December 05, 2020 to March 31, 2021, we used Twitter API[2] to collect tweets directly from Twitter using the COVID-19 related keywords. We filtered the collected tweets based on USA geo-location and English language. We selected only those tweets that had zip-level geo-tag within USA, having 840,000 tweets from April 01 to December 04, 2020. We used TextBlob [3], a natural language processing library, in order to find out the sentimental polarity (i.e., *positive*, *neutral*, or *negative)* of tweets as well as tokenizing keywords.

For the COVID-19 data during the time between April 2020 to March 2021, we used two sources, i.e., Johns Hopkins University Center for Systems Science and Engineering (JHU CSSE) github repository[4] and the New York Times github repository[5]. We selected the data that was based on USA geo-location.

[1]https://tweetsets.library.gwu.edu/
[2]https://developer.twitter.com/en/docs/twitter-api
[3]https://textblob.readthedocs.io/en/dev/
[4] https://github.com/CSSEGISandData/COVID-19
[5]https://github.com/nytimes/covid-19-data

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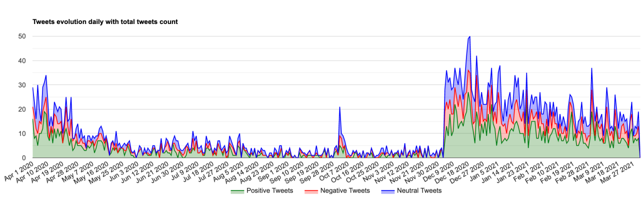

Figure 1: The one year time-line view of tweets' sentimental polarity for Santa Clara country in California, USA.

3 THE VISUALIZATION SYSTEM

In our visualization system, we use a number of interactive visualizations to help users in exploring tweet data between April 2020 to March 2021 targeting COVID-19. The resulting visualizations are based on USA geo-location and can be filtered based on state or county level. This gives us the option to compare between two or more areas in order to see how people reacted to COVID-19 at the same time or over the period of a timeline.

We provide sentimental polarity of tweets over the time (see Figure 1) through a time-based stack area chart, where green color represents tweets with *positive* polarity, blue color represent tweets with *neutral* polarity, and red color represents tweets with *negative* polarity. The resulting visualization can be filtered based on a number of options, e.g., state, country, sentimental polarity, and time interval. Clicking on any sentimental polarity at any time point shows further details through a tool-tip, e.g., the number of tweets on that day with the underlying polarity, total tweets on that day, new COVID-19 cases on that day, and new deaths due to COVID-19 on that day (see Figure 2). All these information are calculated based on the selected geo-location in the resulting visualization.

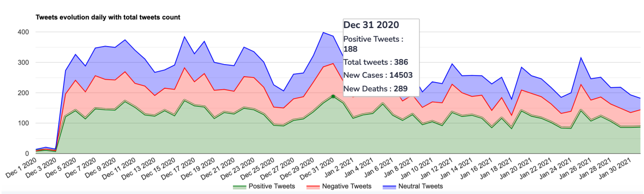

Figure 2: Two months' time-line view of tweets' sentimental polarity for Los Angeles country in California, USA with tool-tip.

Showing frequent keywords from people's tweets reveals what people are talking mostly towards the underlying event. We show the keywords frequency evolution over time through two visualizations. In both cases, users can filter the resulting visualization based on geo-location (i.e., county or state level) and time interval (currently, the minimum time difference is one day).

In the first case, we show the most frequent keywords through a stack column chart (see Figure 3). There are two options, i.e., showing the most frequent keywords at each time period or showing the overall most frequent keywords and their evolution over all the selected time interval (e.g., Figure 3). Users can select the number of frequent keywords, time interval, and geo-location (state or county). Mouse hovering over a particular bar brings a tool-tip with further details about each keyword.

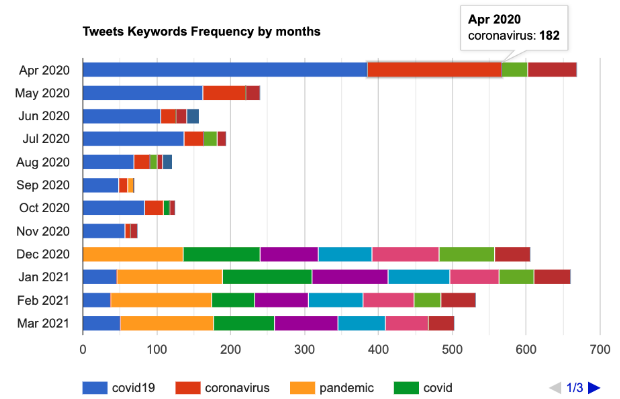

Figure 3: The ten most frequent keywords over the time period of one year (showing per month) for Santa Clara county in California, USA.

In the second case, we show the keywords evolution over time through a continuous stacked area chart (see Figure 4). By default, we show the top twenty keywords and their evolution over time (Figure 3); however, users can filter the visualization based on geo-location (county and state), number of most frequent keywords, and time interval. Again, clicking on a particular keyword line brings a tool-tip with further details. Such a continuous time-based visualization gives us an indication when there were sparks in people's tweets, most probably due to sparks in the COVID-19 cases, and what were the main topics during that time.

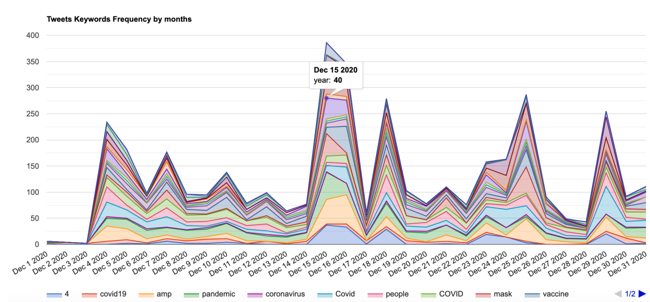

Figure 4: The top twenty frequent keywords evolution over the period of December 2020 for San Francisco county in California, USA.

In the future, we plan to include visualizations in our system based on COVID-19 data as well. Further, we intend to find the relation between people's tweets and spreading of COVID-19 cases based on geo-location. We would like to investigate whether we can predict rise of COVID-19 cases based on analyzing people's tweets in a specific geo-location. Finally, we plan to do user study in order to see the usability of our visualization system.

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